

Virtual multi-staining in a single-section view for renal pathology using generative adversarial networks

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ABSTRACT

Sections stained in periodic acid-Schiff (PAS), periodic acid-methenamine silver (PAM), hematoxylin and eosin (H&E), and Masson's trichrome (MT) stain with minimal morphological discordance are helpful for pathological diagnosis in renal biopsy. Here, we propose an artificial intelligence-based re-stainer called PPHM-GAN (PAS, PAM, H&E, and MT-generative adversarial networks) with multi-stain to multi-stain transformation capability. We trained three GAN models on 512×512 -pixel patches from 26 training cases. The model with the best transformation quality was selected for each pair of stain transformations by human evaluation. Frechet inception distances, peak signal-to-noise ratio, structural similarity index measure, contrast structural similarity, and newly introduced domain shift inception score were calculated as auxiliary quality metrics. We validated the diagnostic utility using 5120×5120 patches of ten validation cases for major glomerular and interstitial abnormalities. Transformed stains were sometimes superior to original stains for the recognition of crescent formation, mesangial hypercellularity, glomerular sclerosis, interstitial lesions, or arteriosclerosis. 23 of 24 glomeruli (95.83 %) from 9 additional validation cases transformed to PAM, PAS, or MT facilitated recognition of crescent formation. Stain transformations to PAM ($p = 4.0E-11$) and transformations from H&E ($p = 4.8E-9$) most improved crescent formation recognition. PPHM-GAN maximizes information from a given section by providing several stains in a virtual single-section view, and may change the staining and diagnostic strategy.

1. Introduction

Renal biopsy is the gold standard technique for renal disease diagnosis. Diagnosis from renal biopsy specimens is important for taking clinical measures and preventing renal impairment [1]. In addition to standard H&E-stained sections, to elucidate glomerular structures such as the mesangial area, basement membrane, or interstitial fibrosis, specialized staining techniques such as periodic acid-Schiff (PAS), periodic acid-methenamine silver (PAM), and Masson's trichrome (MT) are prepared [2]. Making serial sections is a common practice for observing the closest views to the neighboring sections. However, even if these four stains are performed on serially cut sections, the first and last sections will be separated by 10–20 μm . While the intervals between sections may introduce only minor variations in the images, they can lead to significant pathological differences. The Oxford Classification of IgA nephropathy includes a crescent score (C0, C1, or C2), where even the presence of a single additional crescent can upgrade the score from

C0 to C1, signifying a higher risk of poor renal outcomes [3]. In cases where a critical finding is only visible in one particular stain, which may not be as effective in highlighting that finding as other stains, maximizing the information derived from that specific stain becomes essential to ensure the finding is accurately identified.

A specialized deep neural model called PAS, PAM, H&E, and MT generative adversarial network (PPHM-GAN) is developed for multi-stain transformation to mitigate these problems and validated for its diagnostic utility by pathological evaluation.

AI featuring deep neural networks has been proven to be powerful in digital pathology areas such as tumor detection [4], predicting micro-satellite instability status [5], and synthesis of diagnostic images [6]. Generative adversarial networks (GAN) [7] with a StarGAN [8] backbone was adopted for the stain-transforming AI. GAN is an artificial synthesis framework with two neural networks: the generator and the discriminator. The generator is a generating network that can synthesize data from inputs such as random numbers [6,7,9,10], pairs of images

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[11], and unpaired images [12]. A simple encoder-decoder model trained with GAN can generate images with low latency and sufficient throughputs for real time synthesis and whole slide image transformation, providing an advantage over recent diffusion models [12, 13]. Measuring AI performance is important for comparison but GAN has no natural metrics such as accuracy in supervised classification tasks for performance measurement. Instead, human evaluation of GAN-synthesized images is a standard and reliable practice [6,11] as ultimately the pragmatic value of fake data is determined by humans. The Frechet inception distance (FID), peak signal-to-noise ratio (PSNR), and structural similarity index measure (SSIM) are often used as an AI-based visual quality metric and reconstruction quality metrics, respectively [6,9,10,13,14]. Several studies have employed contrast structural similarity (CSS) [15,16] and Wasserstein distances [17,18]. In this work, we introduced a variant of the domain shift score [17], called the Domain Shift Inception Score (DSIS), which measures the Wasserstein distances between feature maps of a pre-trained InceptionV3. We utilized FID, DSIS, and CSS to measure differences between stains before and after transformation, and PSNR and SSIM to assess reconstruction quality within the same stain after it was transformed twice back to the original stain.

CycleGAN [12] was proposed as a GAN framework for image-to-image transformation for unpaired images of two domains. CycleGAN transforms an image from one domain to another while preserving its underlying structure, with the help of cycle consistency loss calculated by the difference between the original domain images and the images transformed twice back to the original domain. The cycle consistency loss forces the networks to work as a one-to-one mapping between the two domains, while the GAN mechanism maintains the plausibility of the fake images. As a use case, De Bel et al. used a modified CycleGAN to standardize PAS-stained renal tissue sections from different centers [19], and you can find several more in the literature [20,21]. While CycleGAN requires the same number of generators and discriminators as transformation pairs for multiple-domain transformation, StarGAN was proposed to use a single generator and discriminator for multiple-domain transformation, then improving efficiency [8]. The StarGAN approach is computationally efficient, as the training complexity of CycleGAN increases quadratically with the number of stains. Specifically, $\frac{1}{2}n(n-1)$ distinct training is required, where n represents the number of stains. The number of stains could be expanded from four in our study to include additional stain, such as CD31/CD34 immunohistochemistry for identifying vasculature, CD68 immunohistochemistry for endocapillary hypercellularity, and C4d for detecting antibody-mediated rejection. Data availability in normal facilities and a computationally tractable framework are important as the AI model should be run in each facility to accommodate data diversity between facilities. AI models requiring specialized data such as auto-fluorescence [22,23], or bleaching and re-staining [24], or fine-grained segmentation masks [15] were not considered feasible. For the above reasons, the StarGAN framework was appealing as it requires only unpaired images of each domain and can perform multi-stain to multi-stain transformation with less computation than CycleGAN.

In this paper, a virtual re-staining AI platform, PPHM-GAN is proposed as a multi-stain to multi-stain transformation platform. The unique aspect of this approach is that the pathologists are aware of the data authenticity, distinguishing between real and artificially transformed sections. The diagnostic utility of PPHM-GAN is proven if any artificially transformed section enhances the identification of a certain finding essential in renal disease diagnosis. Tools for local training are provided including PyTorch training codes and FID/DSIS/CSS/PSNR/SSIM with two-dimensional (2D) multidimensional scaling (MDS) visualization before human selection. Several diagnostic findings were screened by a single pathologist. Thereafter, the subsequent in-depth assessment of a targeted finding was conducted by multiple pathologists with statistical analysis.

2. Methods

The entire study design is illustrated in Fig. 1. The PyTorch implementation of PPHM-GAN is available at <https://github.com/enigmanx20/PPHM-GAN>.

2.1. Case selection and whole-slide image acquisition

A total of 26 renal biopsy cases for training and 19 cases for validation were enrolled that were sampled between 2018 and 2020 at the University of Yamanashi Hospital. All cases were Asian. The included cases were chosen to include a variety of renal diseases and pathological abnormalities. Biopsies for tumor diagnosis or transplant cases were excluded. The training and validation cases were not strictly randomized due to the diversity of diseases. The clinical information including age, sex, and primary diagnoses is listed in Supplemental Table 1.

Ten validation cases were used initially to evaluate general renal findings (Case 1–10) and another nine cases were gathered after the enrollment of Case 1–10 to evaluate crescent formation (Cre 1–9). All specimens were formalin-fixed and paraffin-embedded, and stained with PAS, PAM, H&E, and MT using standard protocols. The stained sections were cut in close sequence, though not strictly serial, as sections with holes, scratches, or poor stain quality were discarded. Tissue sections from renal biopsies were stained directly, without any bleaching or re-staining. Whole-slide images (WSIs) of stained sections were acquired using a virtual slide scanner (VS-120, Olympus, Tokyo, Japan) with a 40 $\times$ objective lens (0.17 μ m/pixel) in bright field. No immunofluorescence channel was used. Extra sections from the same paraffin block were available in most cases and used for model selection, as described in the next section.

2.2. Compiling datasets

Five datasets of differently sized image patches were compiled: training set (512 $\times$ 512-pixel or 87 $\times$ 87 μ m patches) and model selection set (2048 $\times$ 2048-pixel or 348 $\times$ 348 μ m patches) from training cases, validation set (512 $\times$ 512-pixel or 87 $\times$ 87 μ m patches) and human validation set (5120 $\times$ 5120-pixel or 870 $\times$ 870 μ m patches) from Case 1–10, and crescent evaluation set (5120 $\times$ 5120-pixel or 870 $\times$ 870 μ m patches) from Cre 1–9.

The training set was composed of 512 $\times$ 512-pixel patches (87 $\times$ 87 μ m) from the WSIs of 26 training cases. A 512 $\times$ 512-pixel window raster scanned the entire slide with a stride of 256 pixels horizontally and vertically. The patch in the window was screened to contain valid tissue using the patch JPEG file size with manually set thresholds before stored in PNG format for model training and validation. In total, 181 586 PAS, 200 481 PAM, 186 800H&E, and 174 302 MT patches made the training set. To choose the trained model with the best quality by human assessment, a model selection set of 2048 $\times$ 2048-pixel patches (348 $\times$ 348 μ m) was prepared by manually cropping representative cortical and medullary regions from the WSIs from different sections of the training cases. The validation set was composed of 512 $\times$ 512-pixel patches from validation case WSIs, which were cropped using the same protocol as for the training set. In total, 53 842 PAS, 59 957 PAM, 63 185H&E, and 60 977 MT patches were enrolled in the validation set. To make the human validation set, MK cropped five representative 5120 $\times$ 5120-pixel patches (870 $\times$ 870 μ m) manually from the WSIs of ten validation cases (Cases 1–10 in Supplemental Table 1), with the majority chosen to include regions of the glomeruli and also at least one patch of the medullary region. To make the crescent evaluation set, between one and five 5120 $\times$ 5120-pixel patches were cropped manually from the WSIs of nine additional validation cases (Cre 1–9). Regions of glomerular crescents were chosen and each crescentic glomerulus was numbered 1 to 24 (Glomerulus 1–24).

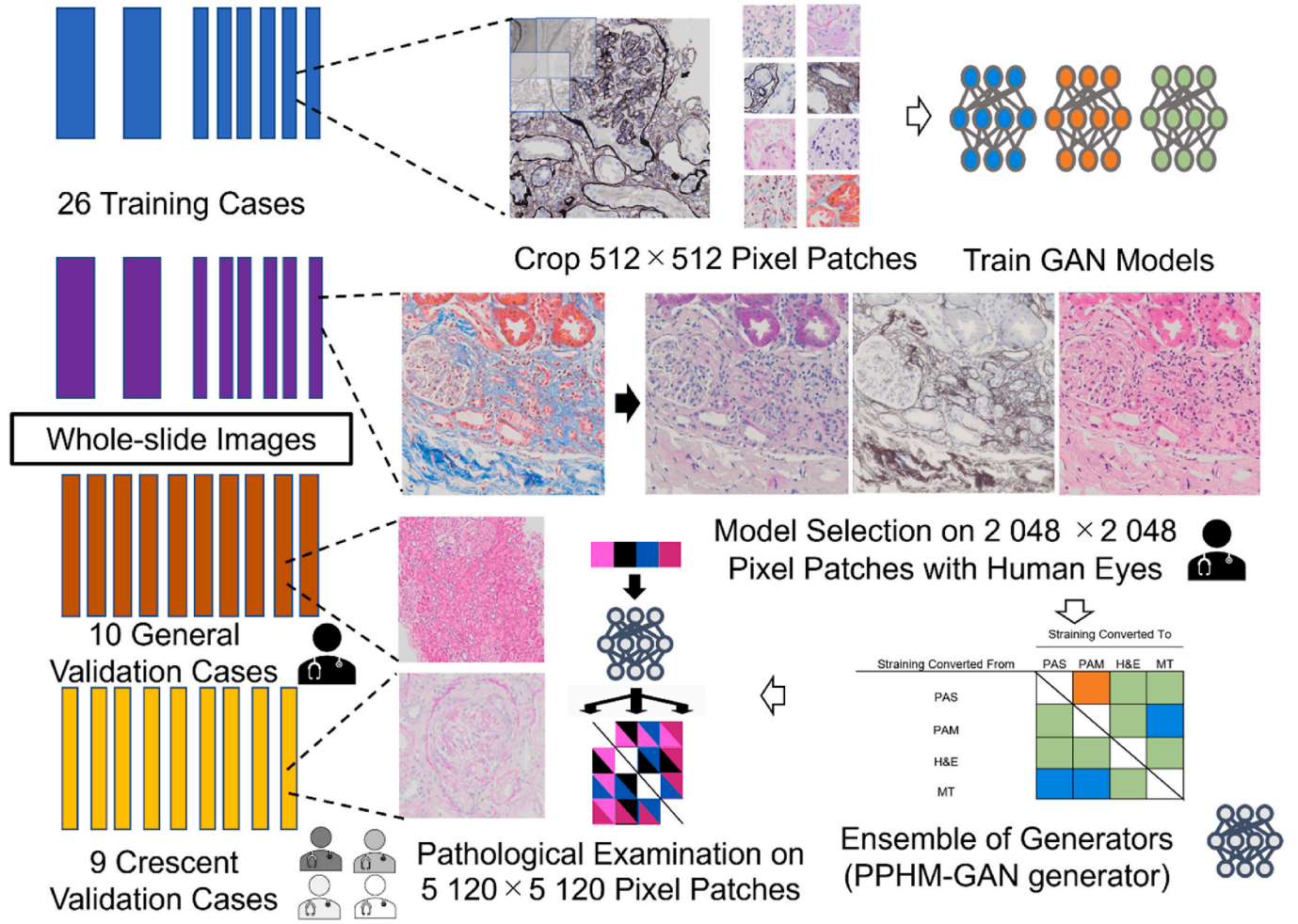


Fig. 1. The overview of the study. Whole-slide images (WSIs) of 26 training, 10 general validation cases, and 9 crescent validation cases were enrolled. The vertical left most bars represent WSIs used in this study. 512×512 -pixel patches were cropped from training and general validation WSIs for GAN training and FID calculation. Three GAN models were trained. A human pathologist evaluated 2048×2048 -pixel patches from another section WSIs of training cases to select the model out of three trained models by the image quality. 5120×5120 -pixel patches were cropped from validation cases and evaluated by four pathologists. WSI: whole slide image, GAN: generative adversarial network.

2.3. Frechet inception distance (FID) and domain shift inception score (DSIS)

Generated images were assessed using the Frechet Inception Distance (FID) [25], defined as the relative distance between groups of images calculated by $(\rho_1 + \rho_2 - 2(\rho_1 \rho_2)^{1/2})^{1/2}$, where μ_i and ρ_i are the mean and covariance matrices of the last pooling layer of InceptionV3 [26] pre-trained on ImageNet [27]. FID measures the similarity between two groups of images. Generally, a lower FID indicates better synthesized images. FID was first proposed as a robust evaluation metric for general images but has been applied successfully to pathological images and used for GAN model evaluation [6]. Although the ImageNet pre-trained InceptionV3 is not optimized for histopathological images, it is comparable with the same model pre-trained on a large H&E dataset in fine-tuning capability on H&E tasks [28]. Considering large stain variations in special stains, ImageNet pre-trained InceptionV3 was adopted for FID backbone instead of an ad hoc model used in Ref. [14]. As pooling more than 10K images from each image group is recommended for a robust FID, 50K images and 30K images were assessed for the training and the validation sets respectively, alongside their output transformed images. The real images were randomly selected from the original 512×512 -pixel images from the dataset, scaled within the range of $(-1.0, 1.0)$ prior to processing with the InceptionV3 network. The outputs of the last pooling layer were

averaged across samples, and the mean and covariance matrices were stored for each channel. Similarly, the fake images were generated using the same original input images as mentioned above. The mean and covariance matrices of each channel were calculated similarly. The FID between the real images and their corresponding transformed images was computed using the previously mentioned formula.

Building on the concept of the domain shift score proposed in Ref. [17], we introduced the domain shift inception score (DSIS). DSIS is calculated as the average of the Wasserstein distances between intermediate feature maps of an InceptionV3 model pre-trained on ImageNet. While [17] used the domain shift score to measure shifts within the same stain as a performance degradation factor, we adopted DSIS to assess color differences between the original and transformed images, similar to the approach in Ref. [18]. This adjustment reflects our focus on human evaluation where domain shifts in the same stain is not as relevant as AI-based assessments. Specifically, DSIS of two images x and y are $DSIS(x, y) = \frac{1}{k} \sum_k WassersteinDistance(\varphi_x^k, \varphi_y^k)$, where φ_x^k is the k th

feature map of image x . We collect four intermediate feature maps after each inception block. Random 1K images were transformed to each stain for training and validation sets. DSIS between real images and the corresponding transformed fake images was assessed three times, and the mean of the repetitions is reported. While FID measures between sets of images, DSIS measures between an image pair before and after

transformation.

2.4. Peak signal-to-noise ratio (PSNR)

PSNR measures reconstruction error in dB by $PSNR = -10 \times \log_{10}(\text{Mean Squared Error})$ over the whole image pixels [13]. Generally, higher PSNR indicates better reconstruction. We assessed 30K images and averaged them for the validation set, alongside their output transformed images. The real images were randomly selected from the original 512×512 -pixel images from the dataset, scaled within the range of $(-1.0, 1.0)$ before being transformed by the PPHM-GAN generator. The images were transformed twice back to the original stain. The PSNR between the real images and their corresponding reconstructed images was computed using the aforementioned formula. In a loop from the original stain to another and back to the original stain, the transformed stain is the informational bottleneck. Therefore, the drops of PSNR during these transformation loops reflect the information loss or the difficulty of the transformation. It is important to note that PSNR is closely related to the cycle consistency loss described in the model training details section, which is optimized during training, with the difference being that PSNR utilizes L2 loss while cycle consistency loss employs L1 loss.

2.5. Structural similarity index measure (SSIM) and contrast structural similarity (CSS)

Structural similarity index measure (SSIM) measures similarity between two images respect to luminance, contrast, and structure. Contrast structural similarity (CSS) is a variant of SSIM, removing the luminance component. Generally, higher SSIM indicate better reconstruction and CSS indicate better structural preservation after stain transformation. Specifically, SSIM of two images x and y is $SSIM(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}$, where μ_x, μ_y are the mean σ_x, σ_y are the standard deviations, σ_{xy} is the covariance, of a sliding window on x and y . C_1, C_2 are the stabilizing constants. Similarly, $CSS(x, y) = \frac{(2\sigma_{xy} + C_3)}{(\sigma_x^2 + \sigma_y^2 + C_3)}$, where C_3 is the stabilizing constant. In our study, SSIM and CSS are calculated using a sliding window of size 11 per channel and averaged over all windows and channels. Stabilizing constants were set as $C_1 = 1E - 4$ and $C_2 = C_3 = 9E - 4$. We assessed SSIM and CSS of 30K images and averaged. The images of the training sets were transformed twice back to the original stain for SSIM in the same manner as PSNR. The original images of the training and validation sets and transformed images are compared for CSS in the same manner as DSIS.

2.6. Stain transformation distance and multidimensional scaling (MDS)

The stain transformation distance matrices between the stains for FIDs, DSIS, CSS, PSNR, and SSIM were defined as follows; From 4×4 matrices of stain pairs, diagonal elements were subtracted along columns after reversing signs for CSS, PSNR, and SSIM matrices. Negative values were set as zeros and were averaged with the transposed matrices to be symmetrical. We visualized the stain transformation distance matrices using multidimensional scaling (MDS), which visualizes points in 2D space according to the distance matrix. Stress-1 values were calculated by $\text{Stress} - 1 = \sqrt{\sum_{i < j} (d_{ij} - \hat{d}_{ij})^2 / \sum_{i < j} d_{ij}^2}$ as the measure of the fitness of good, where d_{ij} is the original distance between element i and j and $\hat{d}_{ij}$ is the embedded distance between element i and j .

2.7. Human evaluation

Five skilled Japanese pathologists (denoted MK, TO, KK, TI, and KM) who routinely diagnose renal biopsy were recruited to independently evaluate the images. MK selected models from the model selection set

and evaluated common renal findings in Cases 1–10. MK also performed organizational tasks such as manually cropping images from WSIs and experimental setup. TO, KK, TI, and KM evaluated 24 glomeruli (Glomerulus 1–24) chosen from nine extra cases (Cre 1–9) for crescent evaluation. They were kept blinded to the original real images and the model selection process until the completion of fake image evaluations.

All evaluators used the same scale to grade the effectiveness of an image or glomerulus for evaluating findings: good (G), acceptable (A), equivocal (Q), and bad (B), corresponding to the numbers 3, 2, 1, and 0 respectively. An image or glomerulus was scored G (good) when it was good for evaluating certain findings and diagnosis was possible with confidence solely from the image. An image or glomerulus was scored A (acceptable) when it was acceptable for evaluation, and the diagnosis was possible solely from the image but with ambiguity. An image or glomerulus was scored Q (equivocal) when the evaluation was equivocal, with the presence of a finding suspected from the section, and/or when the finding could be confirmed by cross-reference to other images of the same glomerulus. An image or glomerulus was scored B (bad) when diagnosis could only be suspected for a certain finding by cross-reference to other stains of the same glomerulus.

MK evaluated the following findings for the human validation set: spikes and holes in the basement membrane, meaning the presence and extent of spikes and holes, which indicate immune complex deposits of membranous nephropathy. crescent formation in glomeruli, meaning the cellular and/or fibrous proliferation along Bowman's capsule, a critical marker of glomerular injury. mesangial hypercellularity, meaning increased number of mesangial cells, which is associated with various forms of glomerulonephritis. glomerular sclerosis, meaning the extent of glomerular scarring or sclerosis, indicative of chronic kidney damage. interstitial inflammation, meaning the degree of inflammation within the renal interstitium. interstitial fibrosis and tubular atrophy, meaning comprehensive evaluation of the extent of interstitial fibrosis and tubular atrophy, which are key indicators of chronic renal injury. And arteriosclerosis, meaning the thickening and hardening of arterial walls within the kidney, which is commonly associated with hypertension and chronic kidney disease. The grades were a comprehensive evaluation of the human validation set.

TO, KK, TI, and KM evaluated crescent formation for the crescent evaluation set by reporting grades of original images or glomeruli as one of G, A, Q, or B. Subsequently, they evaluated the transformed images and reported the grading differences in integer values compared to the original staining. Cohen's quadratic kappa values were calculated to measure inter-grader agreement among TO, KK, TI, and KM. The ratios of upgraded glomeruli to total glomeruli were recorded for each transformed stain. Fisher's exact test was performed, assuming the null hypothesis of no association between the transformation and upgrades.

2.8. Model architecture

The deep generator and discriminator neural nets comprise stacks of two dimensional convolution of output channels n , kernel size k , and stride m (denotes $\text{Conv}(n, k, m)$) with and without bias (w/B and w/oB, respectively); transposed two dimensional convolution of output channels n , kernel size k , and stride m (denotes $\text{TConv}(n, k, m)$) with and without bias (w/B and w/oB, respectively); upsampling of kernel size k (denotes $\text{Up}(2)$); average pooling of kernel size k (denotes $\text{AvgPool}(k)$); instance normalization (IN); rectified linear unit (ReLU); leaky ReLU of negative slope 0.01; and tanh activation. Before each Conv layer, the channels are padded to be the same image size. Residual blocks containing layers such as $\text{Conv2d}(n, k, m)$ and ReLU with skip connection x are denoted as $\text{Res}[\text{Conv2d}(n, k, m), \text{ReLU}, +x]$, where x is the residual term from the input of the residual block.

The generator series is as follows: $\text{Conv}(64, 3, 1)$ w/oB, $\text{Conv}(64, 3, 1)$ w/oB, IN, ReLU, $\text{Conv}(128, 5, 2)$, $\text{AvgPool}(2)$, IN, ReLU, $\text{Conv}(256, 5, 2)$ w/oB, $\text{AvgPool}(2)$, IN, ReLU, $\text{Res}[\text{Conv}(256, 3, 1)$ w/B, IN, ReLU, $\text{Conv}(256, 3, 1)$ w/B, IN $+x$] $\times 6$, $\text{TConv}(128, 5, 2)$, IN, ReLU, $\text{TConv}(64,$

5, 2), IN, ReLU, Conv(3, 7, 1), and Tanh. The discriminator consists of a series of AvgPool (2), Conv(64, 5, 2) w/B, Leaky ReLU, AvgPool(2), Conv(128, 5, 2) w/B, leaky ReLU, AvgPool(2), Conv(256, 5, 2) w/B, leaky ReLU, AvgPool(2), Conv(512, 5, 2) w/B, leaky ReLU, AvgPool(2), Conv(1024, 5, 2) w/B, leaky ReLU, AvgPool(2), Conv(2048, 5, 2) w/B, leaky ReLU, AvgPool(2), Conv(4096, 5, 2) w/B, leaky ReLU, and Conv (5, 8, 1) w/oB. The discriminator 5 output channels are globally average pooled to yield one GAN output and four logit outputs. The model architecture is illustrated in [Supplemental Fig. 1](#).

2.9. Defining loss functions and regularizations

We combined two types of GAN losses, classification loss, and cycle consistency loss with regularization techniques for training the discriminator and generator. Specifically, the discriminator term of hinge GAN loss is $L_{hinge}^D = E[\max(0, 1 - D_{real, GAN})] + E[\max(0, 1 + D_{fake, GAN})]$ and the generator term is $L_{hinge}^G = -E[D_{fake, GAN}]$. Where $E[x]$ is the expectation function, $D_{real, GAN} = D(x)$ is the discriminator GAN output of the real images x , $D_{fake, GAN} = D(G(x, c))$ is the discriminator GAN output of the fake images generated by the generator from real images x , and c is the stain code in $\{0, 1, 2, 3\}$ corresponding PAS, PAM, H&E, and MT that prompts the stain transforming to. In practice, $E[x]$ calculates the mean of a minibatch and c is encoded as a one-hot vector before channel-wisely concatenated with the original image. The discriminator term of vanilla GAN loss is $L_{vanilla}^D = -E[\log(D_{real, GAN})] - E[\log(1 - D_{fake, GAN})]$ and the generator term is $L_{vanilla}^G = E[\log(1 - D_{fake, GAN})]$.

Classification loss of the discriminator is $L_{class}^D = E[\text{CrossEntropy}(D_{real, class}, c)]$, where CrossEntropy is the cross-entropy loss taking $D_{real, class}$ as the input and c as the target, $D_{real, class}$ is the discriminator logit outputs. Similarly, classification loss of the generator is $L_{class}^G = E[\text{CrossEntropy}(D_{fake, class}, c)]$.

Cycle consistency loss term of the generator is $L_{cycle}^G = E[\|x - G(G(x, c'), c)\|_1]$. Where c' is the stain code transforming to and c is the original stain code as the real image x .

We applied spectral normalization and orthogonal regularization following [9] to hinge GAN loss, and R1 gradient penalty [29] to vanilla GAN loss to stabilize the training following [30]. Specifically, for each Conv and TConv module of the generator and discriminator the weight matrix is spectrally normalized by its maximum singular value. Namely, $W_{SN} = W/\sigma_W$ is applied when calculating the forward pass. Where W is

the original weight matrix and σ_W is its maximum singular value. The orthogonal regularization term is $OR(W) = \beta \|W^T W (1 - I)\|^2$, where β is the tuning hyperparameter, 1 denotes the matrix filled with 1.0, and I is the identity matrix. R1 gradient penalty term is $R_1 = \gamma E[\sum (\nabla D_{real})^2]$. Where γ is the tuning hyperparameter and ∇D_{real} is the gradients of the discriminator weights respect to the real input.

Two versions of total loss TL_{hinge} and $TL_{vanilla}$ are defined as the following:

$$TL_{hinge}^D = L_{hinge}^D + \lambda_{class} L_{class}^D + \sum_D OR(W_D)$$

$$TL_{hinge}^G = L_{hinge}^G + \lambda_{class} L_{class}^G + \lambda_{cycle} L_{cycle}^G + \sum_G OR(W_G)$$

$$TL_{vanilla}^D = L_{vanilla}^D + \lambda_{class} L_{class}^D + R_1$$

$$TL_{vanilla}^G = L_{vanilla}^G + \lambda_{class} L_{class}^G + \lambda_{cycle} L_{cycle}^G$$

where W_D and W_G are weight matrices in the discriminator and generator, respectively. Throughout the study $\beta = 1E-4$, $\gamma = 0.5$, $\lambda_{class} = 1.0$, and $\lambda_{cycle} = 1.0$ were used.

2.10. Model training details

We trained three models Models 1, 2 and 3 and ensembled. Models 1 and 2 with different random seeds minimized TL_{hinge} while Model 3 minimized $TL_{vanilla}$. Weights of Models 1 and 2 were spectrally normalized. All three models were trained using 20M images (~26.9 epochs) with a batch size of 64. The batch size was reduced to 32 for the last 10M images in Models 1 and 2 to enable FP32 calculations, expecting refinement during the final stages of training. The images were randomly flipped horizontally and scaled in the range $(-1.0, 1.0)$. The Adam optimizer was adopted with no weight decay for both the generator and discriminator. The initial learning rates were $1E-5$ for the generator and $3E-5$ for the discriminator. The learning rates were kept constant during the first 10M images, then were decayed according to cosine scheduling during the last 10M images. Gradients were clipped using the global norm of 10.0. Mixed precision was used in some tensor operations during training and evaluation to shorten calculation time and reduce GPU memory usage. The exponential moving average of the generator and discriminator with a coefficient of 0.99 was saved for stability and used for validation.

Table 1

Frechet inception distances (FIDs) between stains.

	Training				Validation			
	PAS	PAM	H&E	MT	PAS	PAM	H&E	MT
PAS	0.0	144.9	36.2	47.2	0.0	129.2	53.6	52.3
PAM	144.9	0.0	183.8	144.3	129.2	0.0	172.9	134.5
H&E	36.2	183.8	0.0	65.4	53.6	172.9	0.0	54.8
MT	47.2	144.3	65.4	0.0	52.3	134.5	54.8	0.0

Table 2

Frechet inception distances (FIDs) of training and validation sets. The values of the selected models and the ranges of three trained models are shown.

Strain Transformed From	Training				Validation			
	Strain Transformed To							
	PAS	PAM	H&E	MT	PAS	PAM	H&E	MT
PAS	5.53 (5.53–9.52)	22.31 (22.31–39.08)	16.87 (16.04–27.03)	14.46 (13.78–15.73)	7.53	30.06	39.84	18.34
PAM	12.08 (12.08–19.26)	6.14 (6.14–14.17)	30.98 (30.98–39.13)	10.77 (10.77–62.63)	12.61	9.62	66.71	16.98
H&E	20.46 (20.46–28.69)	90.98 (81.31–92.39)	15.49 (13.97–17.95)	22.69 (22.69–68.79)	25.50	90.30	37.84	30.01
MT	16.07 (16.07–24.44)	39.58 (29.62–39.58)	17.97 (15.17–22.05)	5.21 (5.21–93.80)	31.65	40.31	21.52	6.77

The values that match the minimum FID are bolded.

Table 3

Domain shift inception score (DSIS) of training and validation sets. The values of the selected models and the ranges of three trained models are shown.

Strain Transformed From	Training				Validation			
	Strain Transformed To							
	PAS	PAM	H&E	MT	PAS	PAM	H&E	MT
PAS	0.076 (0.076–0.094)	0.155 (0.155–0.173)	0.123 (0.123–0.138)	0.134 (0.134–0.148)	0.075	0.153	0.127	0.133
PAM	0.182 (0.183–0.185)	0.083 (0.083–0.104)	0.201 (0.200–0.207)	0.194 (0.185–0.194)	0.179	0.088	0.196	0.191
H&E	0.100 (0.100–0.117)	0.181 (0.181–0.203)	0.093 (0.093–0.104)	0.131 (0.131–0.156)	0.113	0.177	0.112	0.130
MT	0.152 (0.146–0.152)	0.179 (0.179–0.193)	0.144 (0.144–0.158)	0.083 (0.083–0.164)	0.150	0.183	0.143	0.084

The values that match the minimum DSIS are bolded.

Table 4

Contrast structural similarity (CSS) of training and validation sets. The values of the selected models and the ranges of three trained models are shown.

Strain Transformed From	Training				Validation			
	Strain Transformed To							
	PAS	PAM	H&E	MT	PAS	PAM	H&E	MT
PAS	0.715 (0.677–0.715)	0.523 (0.494–0.580)	0.651 (0.626–0.658)	0.552 (0.498–0.552)	0.705	0.506	0.628	0.526
PAM	0.514 (0.489–0.522)	0.78 (0.753–0.780)	0.496 (0.458–0.509)	0.489 (0.489–0.535)	0.527	0.785	0.517	0.491
H&E	0.680 (0.639–0.680)	0.523 (0.399–0.523)	0.684 (0.683–0.693)	0.595 (0.483–0.595)	0.637	0.534	0.636	0.582
MT	0.495 (0.495–0.518)	0.47 (0.464–0.493)	0.553 (0.511–0.553)	0.727 (0.433–0.727)	0.480	0.446	0.546	0.731

The values that match the maximum CSS are bolded.

2.11. Hardware and software specifications

For PPHM-GAN training and validation, a CentOS server was used with eight A100 40 GB GPUs (NVIDIA, California, USA) and CUDA version 11.3.1, Python 3.8.7, and PyTorch 1.10 installed. For FID calculation and image transformation, the same CentOS server was used. Multidimensional scaling and Cohen's quadratic kappa values were calculated using the Python Scikit-learn (version 1.0.1) package. Fisher's exact test was conducted using the Python Scipy (version 1.7.0) package.

3. Consent to participate and ethics approval

We applied Opt-out method to obtain consent. The requirement for ethical approval was waived by the Institutional Review Board (case number 1998) of the University of Yamanashi.

4. Results

Three models with slight differences (Model 1, 2, and 3) were trained and a model with the best transformation quality on the model selection set for each stain transform was chosen. We call this ensemble the PPHM-GAN generator and this was used in the validation process. In the training phase, the GAN training was stable, and no collapse occurred. The classification and cycle consistency losses tended to decrease during training, while the discriminator adversarial losses maintained similar values. The singular values of the weight matrices for Models 1 and 2 tended to increase. Training each model took approximately 2.5 days. FIDs between training and validation sets of PAS, PAM, H&E, and MT were 9.85, 17.54, 15.87, and 9.67, respectively. FIDs between stains are

listed in Table 1. FIDs were calculated between training set images and the generated images from each model (Table 2). Similarly, DSIS and CSS are listed in Tables 3 and 4, respectively.

The minimum and maximum values of the three models are noted in parentheses. The FIDs and DSIS coinciding with the minimum values, and CSS with the maximum value for each model are bolded, confirming the correlation between FID, DSIS, and CSS and human selection. A notation scheme was used where pseudo-staining X derived from an original staining Y is denoted as XfY.

4.1. Overview of pseudo-staining quality in validation cases

In the validation phase, the PPHM generator FIDs, DSIS, CSS, PSNR, and SSIM were calculated between 512×512 -pixel images in the validation set and their generated images as in training. FIDs, DSIS, and CSS are listed in Tables 2–4, respectively. PSNR and SSIM are listed in Tables 5 and 6, respectively.

Validation FIDs were slightly higher than the training FIDs for training patches in all but PAM to H&E transformation pair. We observed substantial differences in FID scores for H&E between the training and validation sets compared to other stains. However, these differences were not consistently reflected in the DSIS and CSS metrics. The relativistic nature of FID makes interpretation challenging, but this discrepancy might suggest the unstable nature of FID compared to DSIS and CSS.

The MDS stress-1 values were 1.08 for FIDs, 0.61 for DSIS, 0.67 for CSS, 0.59 for PSNR, and 0.75 for SSIM, indicating relatively poor fits overall, with DSIS and PSNR showing better fits compared to others. These poor fits are understandable given that we defined the stain distances as non-negative and symmetrical, but did not account for the

Table 5

Peak signal-to-noise ratio (PSNR) of validation sets.

Intermediate Stain	Stain Transformed From and To			
	PAS	PAM	H&E	MT
PAS	26.10	18.77	23.22	21.93
PAM	24.76	22.10	21.72	21.28
H&E	24.34	15.52	23.10	20.69
MT	22.56	17.64	23.45	23.79
None	28.63	25.21	24.78	26.56

Table 6

Structural similarity index measure (SSIM) of validation sets.

Intermediate Stain	Stain Transformed From and To			
	PAS	PAM	H&E	MT
PAS	0.572	0.537	0.506	0.491
PAM	0.559	0.638	0.515	0.443
H&E	0.528	0.422	0.495	0.474
MT	0.516	0.488	0.527	0.573
None	0.694	0.758	0.609	0.710

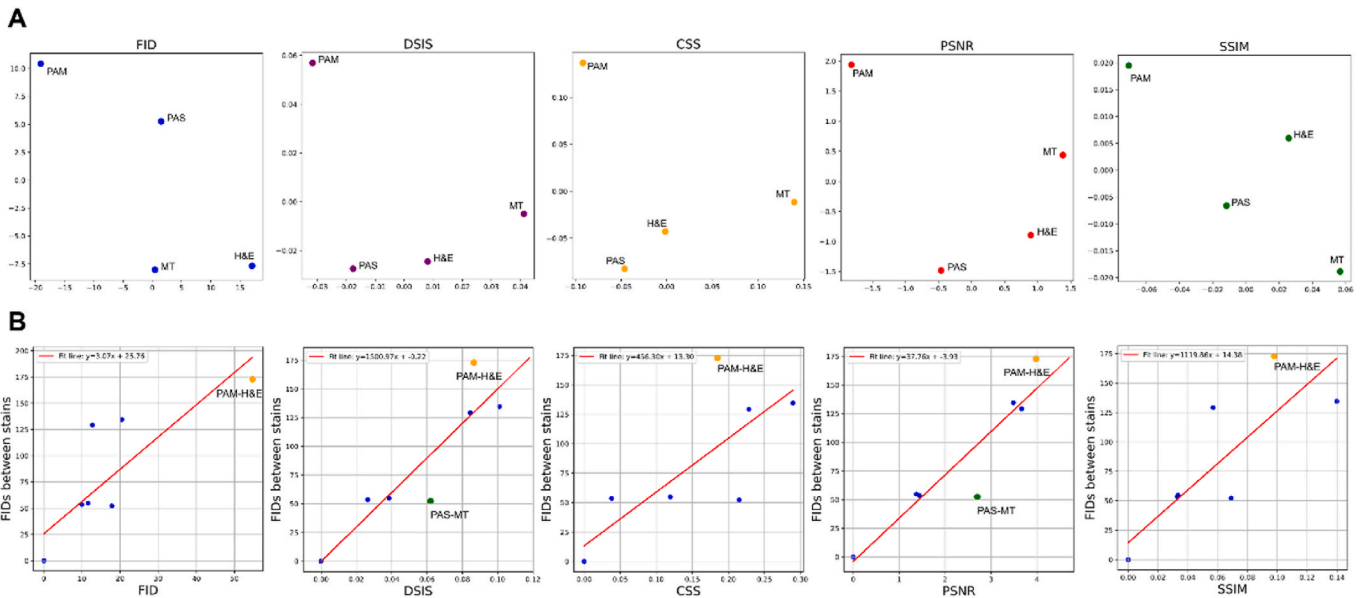


Fig. 2. A: The stain transformation distances between stains for FID, DSIS, CSS, PSNR, and SSIM on the validation set are visualized using MDS. The distances indicate the relative difficulties of stain transformation pairs. H&E and PAM are most distant apart in FID, DSIS, and PSNR, reflecting poor quality transformation between them. Although FID, DSIS and PSNR are distinct metrics, the topologies between stains are similar to each other. B: Linear regression plots of FID, DSIS, CSS, PSNR, and SSIM against FIDs between stains in the validation set. PAM-H&E and PAS-MT pairs are labeled in DSIS and PSNR. MDS: multidimensional scaling, FID: Frechet inception distance, DSIS: domain shift inception score, CSS: contrast structural similarity, PSNR: peak signal-to-noise ratio, SSIM: structural similarity index measure.

triangular inequality. The MDS visualization of stain transformation distances in FID, DSIS, and PSNR shows PAM and H&E transformational pair is the most difficult to transform (Fig. 2A). DSIS visualization is similar in topology as PSNR, whereas the MDS visualization for SSIM and CSS exhibits different topologies and is more challenging to interpret.

The human validation set (Case 1–10) was transformed to each of PAS, PAM, H&E, and MT. The PPHM generator preserved most histological structures or nuclear positions and shapes during pseudo-staining but produced some unfavorable artifacts (Case 10; Fig. 3A and Case 9; Fig. 3B).

4.2. Correlating stain transformation distances with FIDs between stains

We correlated stain transformation distances computed from Tables 2–6 with FIDs between stains in Table 1, assuming transformation difficulty is determined by stain differences. We obtained correlation coefficients of 0.831 with $R^2 = 0.691$ for FIDs, 0.927 with $R^2 = 0.860$ for DSIS, 0.797 with $R^2 = 0.636$ for CSS, 0.947 with $R^2 = 0.897$ for PSNR, and 0.843 with $R^2 = 0.711$ for SSIM in validation set, indicating meaningful fits overall. We plotted the metrics against FIDs between stains (Fig. 2B). Although DSIS has the same backbone as FID, these linear relationships are non-trivial. PAM-H&E pair is depicted as an outlier with insufficient transformation in DSIS plot and better reconstruction in PSNR plot. Whereas, PAS-MT pair is depicted with overdo domain shift and poor reconstruction in PSNR compared to other stain pairs.

4.3. Evaluation of the PPHM generator reliability for pathological findings in general

The grades for common pathological findings are shown in Table 7. MK observed 21 glomeruli with spikes, 16 with holes, one forming a crescent, 13 with mesangial hypercellularity, and seven with sclerosing. The differences from the original images were reported for the pseudo-

staining pairs. Grades of glomerular findings summarized all the glomeruli examined. Pseudo-staining was mostly equal or inferior to the corresponding original staining, but staining pairs such as PAMfPAS and PASfPAM (bold in Table 7) surpassed the original stains for the evaluation of crescentic formation and mesangial hypercellularity.

H&E staining is superior to other stains only in the evaluation of interstitial inflammation and comparable or inferior to PAM, PAS, and MT in other findings. PAS and PASfPAM or PASfMT performed equally well as H&E for the evaluation of interstitial inflammation.

4.4. Assessing pseudo-staining quality for membranous lesions on validation cases

A representative glomerulus with membranous lesions was chosen and compared for every pseudo-staining pair (Case 2; Fig. 4A). Spikes and holes, common findings in membranous nephropathy, were readily identified in the original PAM (Fig. 4B). These findings were slightly visible in transformed PASfPAM (Fig. 4B), H&EfPAM, and MTfPAM images, consistent with normal staining appearance but not as visible as in the original PAM. Although it is not common practice to evaluate spikes and holes using PAS and MT, they were noticeable in these stains by close comparison with PAM. These possible spikes and holes in PAS and MT were transformed to PAMfPAS (Fig. 4C) and PAMfMT.

4.5. Assessing pseudo-staining quality for crescent formation on validation cases

A representative glomerulus with a crescent was chosen and compared for every pseudo-staining pair (Case 1; Fig. 5). Crescents were more visible in the original PAM and PAMfPAS than in the original PAS or PASfPAM stains. The crescent was visible in both MT (as highlighted red color) and transformed MT images, with, MTfPAS, MTfPAM, and MTfH&E also showing red color in the crescentic region, although this was weak for MTfH&E. Conversely, both transformed and original H&E images were not suitable for crescent evaluation.

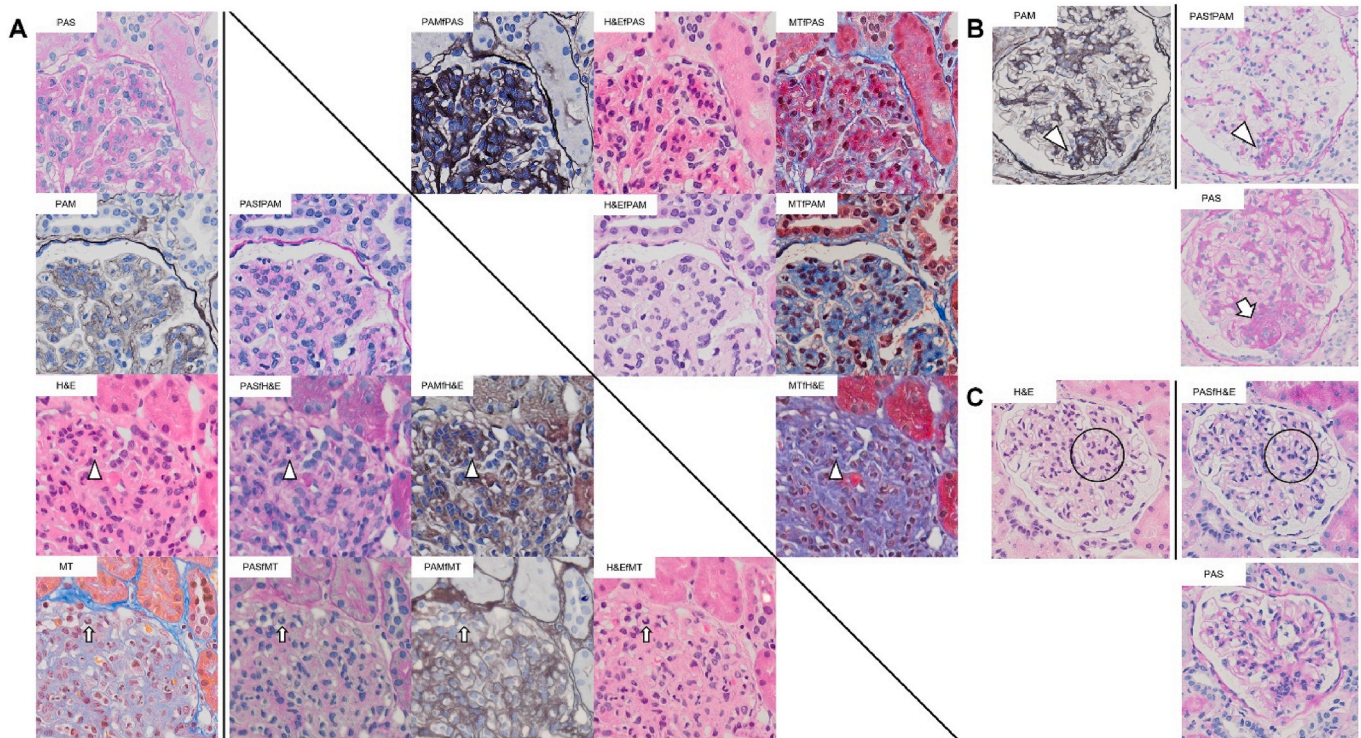


Fig. 3. A: Images of cellular glomerulus and surrounding tubules in the cortex (Case 10). The original PAS, PAM, H&E, and MT images are aligned left of the bar and stain-transformed images right of the bar. The image columns of transformed images are tiled in the order of PASfY, PAMfY, H&EfY, and MTfY from left to right as shown in the captions. A neutrophil is transformed almost as it is in the third row (arrowhead) while a nucleus of a histiocyte loses its characteristic shape after transformation to PAM from MT (arrow). B: The upper panel left to the bar is original PAM, the right upper panel PASfPAM, and the right lower panel original PAS (Case 4). Mesangial hypercellularity, apparent in original PAS staining (arrow), is implied in PASfPAM (arrowhead) while it is ambiguous in original PAM (arrowhead). C: The upper panel left to the bar is original H&E, the right upper panel PASfH&E, and the right lower panel original PAS (Case 9). Mesangial hypercellularity, absent in original PAS staining, is falsely implied in PASfH&E (circle) while it is not clear in original H&E. XfY: pseudo-staining X from Y.

4.6. Crescent evaluation by multiple pathologists

Glomerular crescents are indicative of rapidly progressive glomerulonephritis [31]. Due to their occasional focal and segmental appearance, it is common for a crescent in one glomerulus to be overlooked in a different section during clinical practice. Considering these factors and the insight from the previous section, crescent detection was selected for further comprehensive assessments by multiple pathologists to demonstrate the diagnostic utility of PPHM-GAN.

The grades of pseudo-stained glomeruli (Glomerulus 1–24) were reported as the differences from the original glomeruli averaged between the evaluators. The raw individual evaluations are presented in [Supplemental Table 2](#). Quadratic kappa values between the four graders ranged from 0.32 to 0.65 with a median of 0.55, showing moderate concordance. Some pseudo-stained glomeruli were rated superior to the originally stained ones. Representative glomeruli for which the evaluators more easily recognized crescent formation in pseudo-stain than in the original stain are shown in [Fig. 6](#) (Glomerulus 5 and Glomerulus 15).

Twenty-three out of twenty-four glomeruli (95.83 %) were upgraded in at least one pseudo-staining pair. [Table 8](#) shows the ratios of upgraded glomeruli of each transformation pair and how artificial stain transformation changed crescent recognizability. PAMfPAS was better than original PAS for 14 glomeruli (58.33 %, $p = 8.1E-6$), PASfH&E than original H&E for seven (29.17 %, $p = 0.0094$), PAMfH&E than original H&E for eight (33.33 %, $p = 0.0039$), MTfH&E than original H&E for ten (41.67 %, $p = 0.00060$), and PAMfMT than original MT for 8 (33.33 %, $p = 0.0039$). In general, stain transformations to PAM (PAMfPAS, PAMfH&E, and PAMfMT, $p = 4.0E-11$) and transformations from H&E (H&EfPAS, H&EfPAM, and H&EfMT, $p = 4.8E-9$) most improved crescent formation recognition ([Table 8](#)).

5. Discussion

The single-section view with different stains, if possible, would help renal pathological evaluation but has been difficult to achieve due to technical constraints and required labor. We proposed an artificial multi-stain transformation platform, PPHM-GAN, trained on renal biopsy images. In most cases, pseudo-staining was equal to or inferior to the original staining, and what could be seen was determined largely by the original staining. However, several pathological findings were more evident in the transformed stain than in the original. In several cases, crescent formation was more visible in pseudo-stained PAM images than in the original PAS, H&E, or MT stains. To our knowledge, there have been no studies of high-resolution artificial stain transformation as detailed as $0.17 \mu\text{m}/\text{pixel}$ nor studies of multi-stain to multi-stain transformation showing not only mere visual quality [14,16,18] but also diagnostic utility. PPHM-GAN was trained on WSIs of 26 cases, which is a small cohort compared to other recent AI studies [4,5]. As staining protocols and quality can vary between facilities and virtual slide scanners [19,32], readers are encouraged to train PPHM-GAN (the stains are not necessarily PAS, PAM, H&E, and MT) locally to accommodate in-house data. The objective metrics including FID, DSIS, and CSS were partially correlated with human selection. However, for diagnostic purposes, virtual staining should undergo human quality control, just as traditional staining methods.

There are two major deep neural net approaches for stain transformation [33]: (1) a supervised approach with or without GAN using autofluorescence [22,23] or re-stained sections [24] as the teacher data, and (2) unsupervised CycleGAN-based approach [14–20,34–36]. The term *virtual staining* is justified when the transformed stain closely resembles traditional staining, with minimal structural distortion, thus

Table 7
Common renal pathological findings recognizability. N is the number of glomeruli examined.

Pathological Findings	PAS	PAMPAS	H&EPAS	MT/PAS	PAM	PAS/PAM	H&EPAM	MT/EPAM	H&E	PAS/H&E	PAM/H&E	MT/H&E	MT	PAS/MT	PAM/MT	H&E/MT
Spikes (N = 21)	Q	0	-1	-1	G	-1	-2	-3	B	0	0	0	Q	-1	-1	-1
Holes (N = 16)	Q	0	0	-1	G	-2	-2	-1	B	0	0	0	B	0	0	0
Crescent formation (N = 1)	A	+1	-1	0	G	-1	-3	-1	Q	-1	0	0	A	-1	0	-1
Hypercellularity, mesangial (N = 13)	G	-3	-2	-2	B	+1	0	0	B	+1	0	0	Q	+1	-1	-1
Glomerular sclerosis (N = 7)	A	+1	0	0	A	0	-1	0	Q	0	0	-1	A	-1	+1	-1
Interstitial inflammation	G	-1	-1	-1	A	+1	0	0	G	0	-1	-1	A	+1	-1	-1
Interstitial fibrosis and tubular atrophy	Q	+1	0	0	A	-1	-1	-1	Q	0	-1	-1	G	-2	-1	-3
Arteriosclerosis	A	0	-1	0	A	0	-1	0	Q	0	+1	+1	G	-1	0	-2

XfY: Pseudo-staining X from Y; G: good; A: acceptable; Q: equivocal; B: bad for evaluation. Upgrades are bolded.

perceived as equivalent to applying different stains to the same tissue section. In this sense, PPHM-GAN is a virtual staining technique in the latter category with characteristic multi-stain to multi-stain transformation capability. It is often more labor-intensive to obtain high-quality teacher data using supervised approaches than CycleGAN-based approaches. To replicate the workflow, the unsupervised approach has an advantage over supervised methods [37]. Conversely, CycleGAN-based approaches learn in an unsupervised manner but interpreting the outputs requires additional care. Since GAN models do not take account of the underlying chemicals or histology but optimize only data plausibility, there is a risk that synthesized images are only illusions [22]. Although PPHM-GAN produced several unfavorable artifacts, CycleGAN-based approach is justified from a pragmatic and human-centered standpoint, setting whether the AI helped human experts as the ultimate performance measure.

Theoretically, the mapping of CycleGAN/StarGAN to a different stain is arbitrary, as the cycle consistency loss only ensures a one-to-one mapping, as discussed in Refs. [15,34]. This freedom can lead to two types of hallucinations: color hallucination and structural hallucination. A color hallucination, as discussed in Ref. [22], involves the switching of colors between different structures, often caused by the discriminator's small receptive field. The insufficient domain shift observed in PAM-HE transformations (Figs. 3–5) by PPHM-GAN might also be categorized in color hallucination. Structural hallucinations are unfavorable structural deformation. Although small in deterministic CycleGAN/StarGAN but they become apparent when the model incorporates stochastic inputs allowing high transformation capacity. In our preliminary experiments with StarGANv2 [30], we observed changes in nuclear positions, numbers, sizes, and deformations in tubular contours (results not shown), which are unacceptable for virtual staining, as cautioned in Ref. [36]. High-capacity models, such as StarGANv2, which leverages adaptive instance normalization [10], or diffusion models [13], require additional techniques to simulate virtual staining.

HistoStarGAN [34] attempted to control mapping to preserve glomerular segmentation, but other structures are not necessarily maintained. Similarly, Liu et al. introduced SSIM loss and nuclear segmentation masks as expert knowledge [15]. Classical CycleGAN/StarGAN can achieve virtual staining primarily due to their low transformation capabilities and the use of residual connections that help maintain quasi-identity mapping. For instance, while a 90-degree rotation is a valid one-to-one mapping, CycleGAN does not learn such mappings unless specialized losses or constraints are incorporated into the model architecture.

Based on the observations, we implemented several heuristics to prevent color and structural hallucinations. One approach involved expanding the discriminator's receptive field to capture global image features, reducing the risk of color switching. This adjustment allowed the discriminator to better distinguish between real and fake images from the entire image. We also adopted an ensemble model strategy to mitigate the instability of stain transformations using CycleGAN/StarGAN, which can result in varying quality across different runs [35]. Given that humans can discern trivial color hallucinations, we selected the best model for each stain transformation pair from three models. While model ensemble is more commonly used in supervised tasks [38] and large language models [39], we found that they also offer benefits in our context. This approach allowed us to avoid relying solely on CycleGAN's one-to-one transformations, even though CycleGAN often serves as a strong baseline compared to StarGAN [18,36]. The high-resolution images used in our study demanded significant computational resources. A rough estimate suggests that training 16 CycleGAN pairs (including X2X transformations) would take ~20 days. In contrast, with the same computational resources, we could train three or more StarGAN-like models to choose from.

We could not compare various models systematically as we designed the once and for all human evaluation as the endpoint. Instead, we aligned human evaluation and objective metrics including FID, DSIS,

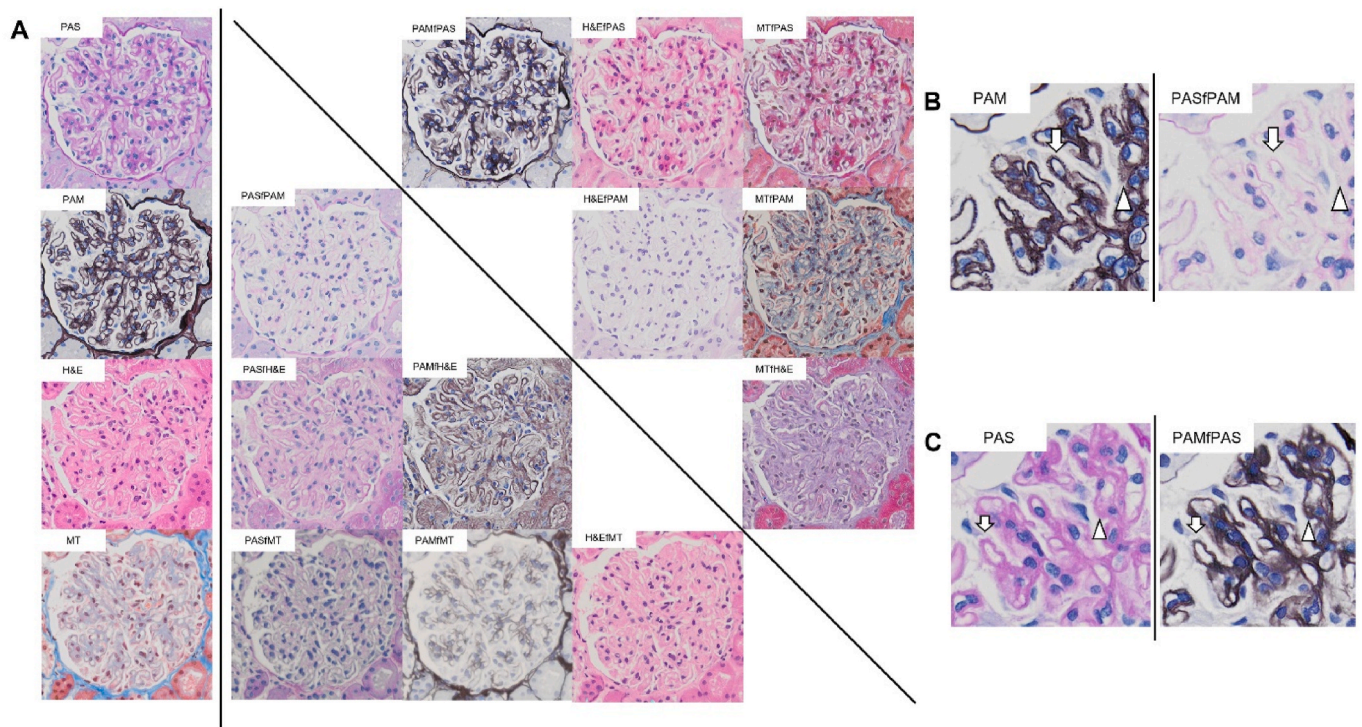


Fig. 4. A representative glomerulus with spikes and holes (Case 2). A: The original PAS, PAM, H&E, and MT images are aligned left of the bar and stain-transformed images right of the bar. The image columns of transformed images are tiled in the order of PASfY, PAMfY, H&EfY, and MTfY from left to right as shown in the captions. B: The panel left to the bar shows the original glomerulus in PAM and the right the transformed glomerulus in PAS (PASfPAM). Spikes (arrow) and holes (arrowhead) are obscured and barely noticeable in PASfPAM while apparent in the original PAM. C: The panel left to the bar shows the original glomerulus in PAS and the right the transformed glomerulus in PAM (PAMfPAS). The spike-like woolly contour (arrow) and hole-like structure (arrowhead) have been preserved. XfY: pseudo-staining X from Y.

CSS, PSNR, and SSIM. Our results suggested transformation difficulties represented by stain transformation distances correlate with stain differences in FID. This is explained that PAM-H&E requires high transformation burden in PPHM-GAN generator as InceptionV3 encodes the same image inputs in distant feature maps. This might explain why CycleGAN that has each model for each transformation is often better than StarGAN that has only single model for every transformation [36]. Style-guided normalization by Y Lin et al. [16] and the style guided blocks by Gual et al. [14] were proposed as the techniques to reduce the transformation burden of the single model approach. Our model may not be state-of-the-art, but the heuristics employed to suppress hallucinations and stain transformation distance analysis are adaptable to other techniques.

The linear relationship observed between FID, DSIS, CSS, PSNR, SSIM, and the FIDs between stains (Fig. 2B) suggests that multi-stain to multi-stain transformation has a technical advantage over single-stain to single stain transformation. Considering their meaning, any single metric cannot capture virtual staining quality. FID overlooks the structural hallucinations as long as the transformed image quality is good. DSIS takes minimal and CSS, PSNR, and SSIM take maximum values when the network learns trivial identity mapping. It's difficult to determine the absolutely optimal values of DSIS, CSS, PSNR, and SSIM. However, we might say which stain pair is insufficiently transformed in comparison with other stain pairs. In practice of StarGAN-like multi-stain to multi-stain model training, the regression lines are expected to show better fits in DSIS and PSNR, and become steeper in FID, CSS, PSNR, and SSIM as the model improves. If a particular stain pair is identified as insufficiently transformed, either through human evaluation or as an outlier in regression analysis, training a CycleGAN-like one-to-one model on that pair would improve the overall model. To further solidify the linear relationships and the interpretability of these metrics, we plan to conduct technical studies in the future.

PPHM-GAN transforms a large diagnostic renal biopsy image into several stains almost instantaneously and once trained, it incurs no extra costs or labor. As such, the superiority of some pseudo-staining pairs for some findings was demonstrated. The results suggest we may change the sectioning and staining habits in the era of AI-aided digital pathology. The PPHM-GAN platform may require fewer sections of H&E than PAM, PAS, and MT as other stains can transform to H&E. Specifically, in a case where a patient presents with hematuria and IgA nephritis is suspected, only PAS, PAM, and MT stains might be prepared to evaluate on Oxford classification. H&E sections could be virtually generated from the other stains. Conversely, in a case with distinct clinical presentation suggesting tubulointerstitial nephritis, H&E sections are prioritized.

De Haan et al. explored deep learning for transforming H&E-stained sections into other special stains using a two-tiered GAN model [22]: A GAN model which transforms autofluorescence images to H&E and another GAN model which transforms H&E to special stains. Despite the ease and cost-effectiveness of H&E stains, they didn't thoroughly analyze the unique insights only special stains could provide. The PPHM-GAN offers an unbiased mutual multi-stain transformation approach, examining the unique features each stain can highlight with human assistance. Contrary to De Haan et al.'s pipeline, the PPHM-GAN platform suggests, at least for crescent evaluation, starting with special staining sections, then transforming them into H&E.

This study has several limitations: (1) Some pseudo-staining pairs, including PAMfH&E, produced lower quality images than others such as PASfPAM.; (2) the validation cohort was small and had low disease diversity, and the pathological findings examined were not exhaustive; and (3) the results of this study, in part rely on subjective human evaluation. Limitation (1) could be solved by choosing cycleGAN-like one-to-one transformation with batch normalization for the low-quality pairs suggested in Refs. [18,36]. The MDS visualization of stain transformation distances and regression analysis with stain differences in FID

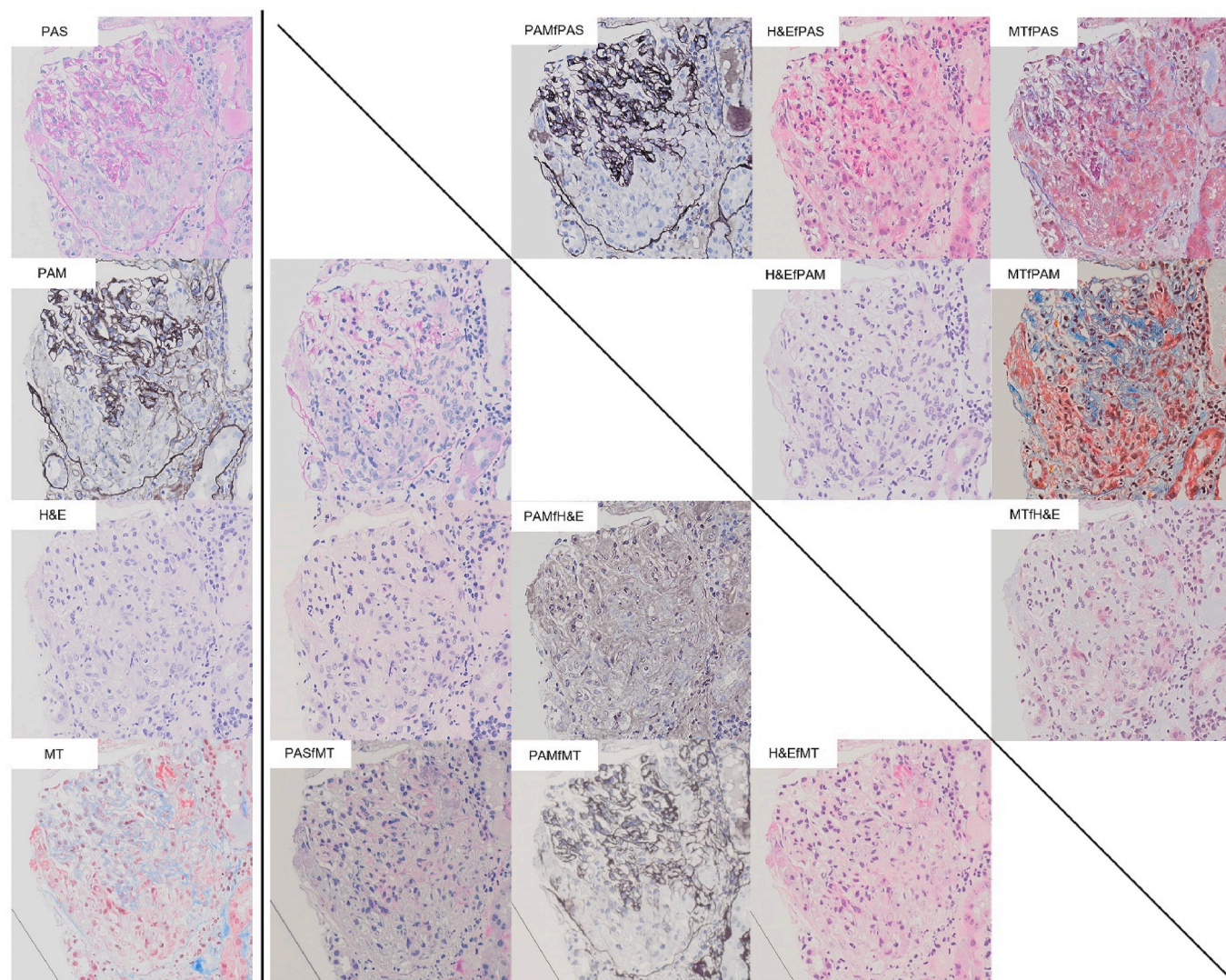


Fig. 5. A representative glomerulus with a crescent (Case 1). The original PAS, PAM, H&E, and MT images are aligned left of the bar and stain-transformed images right of the bar. The image columns of transformed images are tiled in the order of PASfY, PAMfY, H&EfY, and MTfY from left to right as shown in the captions. The crescent formation is more visible in PAMfPAS than the original PAS. The crescent formation is also recognizable in original MT and transformed MT images.

aid in identifying low-quality stain pairs that necessitate further intensive training. As for (2), we had a limited number of renal biopsy cases with uncommon findings available. Our study aims to demonstrate the proof of concept for utilizing generative AI in renal pathology. We plan to conduct a future multi-institutional study to further validate and broaden the applicability of our findings. Limitation (3) is often the case when assessing staining quality. Pathologists usually verify staining quality by confirming histologic components are stained correctly with sufficient contrast. Additionally, pathologists may have their own preferences. The choice of models by another pathologist in the evaluators than MK may lead to different results. Automated stain quality control is an interesting research field in digital and AI pathology, but we have currently no convenient tools available. Validation by supervised AI models [14,17,19,34,35] is considered in future studies. However, caution is needed, as most studies focus on glomerular segmentation, which is one of non-diagnostic tasks in renal pathology. To accurately evaluate the findings in Table 7 with AI models, distinct models for each finding are necessary. To this end, human experts still play a pivotal role in such scenario as expert pathologist annotation is important to develop such models [40].

In summary, we proposed a GAN-based AI platform, PPHM-GAN, which transforms high-resolution images of multiple stains into

differently stained images. The PPHM-GAN generator provides a virtual single-section view of different stains and highlights several findings essential in renal pathology, which contributes to correct diagnosis and an elaborate staining strategy.

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Data availability

The data used in the current study are not publicly available due to the limited data usage requirements for the informed consent of this study but are available from the corresponding author on reasonable request and subject to ethical approval.

During the preparation of this work the authors used Grammarly and ChatGPT in order to correct misspellings, grammar, and sentence structures for improved English writing. After using this tool/service, the authors reviewed and edited the content as needed and takes full responsibility for the content of the publication.

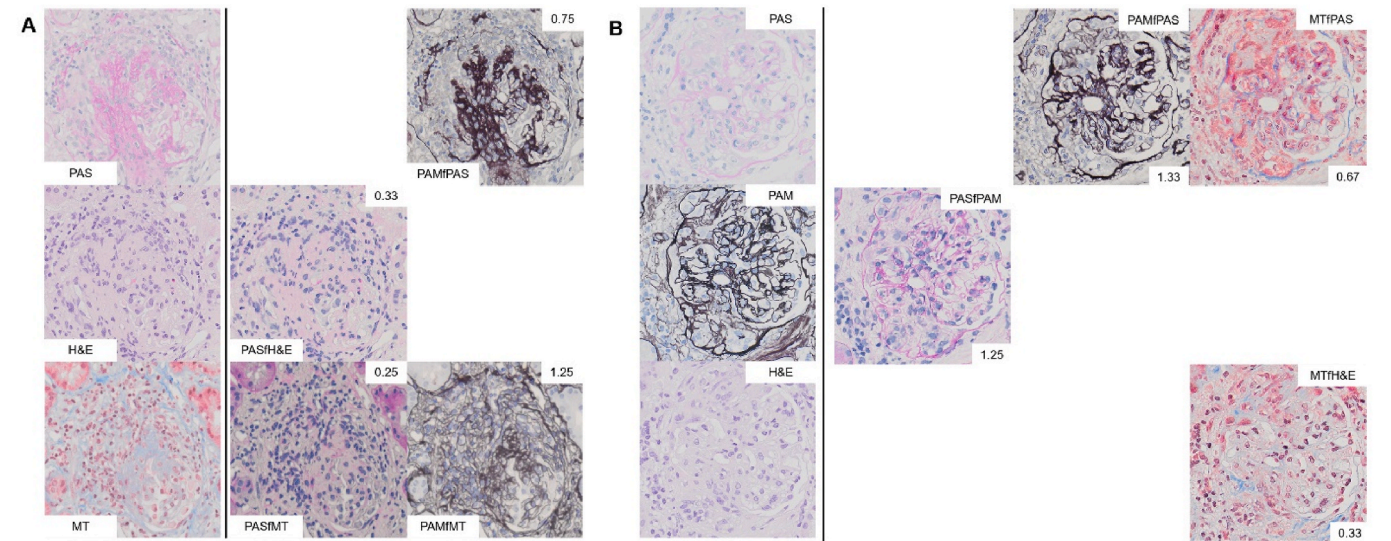


Fig. 6. Representative glomeruli upgraded for recognizing crescent formation in the pseudo-stained images by pathologists' evaluation. The upgrades averaged over graders are captioned in each pseudo-staining panel. A: The original PAS, H&E, and MT images are aligned left of the bar and stain-transformed images right of the bar. The image columns of transformed images are tiled in the order of PASfY and PAMfY from left to right as shown in the captions (Glomerulus 5 of Cre 1). B: The original PAS, H&E, and MT images are aligned left to the bar, and stain-transformed images right to the bar. The image columns of transformed images are tiled in the order of PASfY, PAMfY, and MTfY from left to right as shown in the captions (Glomerulus 15 of Cre 6). XfY: pseudo-staining X from Y.

Table 8

The ratios (%) of upgraded glomeruli in recognizing crescent formation. The right most column, from top to bottom, represents XsfPAS, XsfPAM, XsfH&E, XsfMT. The bottom row, from left to right, represents PASfYs, PAMfYs, H&EfYs, and MTfYs.

Strain Transformed From	Strain Transformed To				
	PAS	PAM	H&E	MT	
PAS	–	58.33***	0.00	4.17	20.83***
PAM	4.17	–	0.00	0.00	1.39
H&E	29.17*	33.33**	–	41.67**	34.72***
MT	12.50	33.33**	0.00	–	15.28**
	15.28**	41.67***	0.00	15.28**	

XfY: Pseudo-staining X from Y; *p < 0.05, **p < 0.005, ***p < 0.0005; The maximum values in the right most column and the bottom row are bolded.

CRediT authorship contribution statement

Masataka Kawai: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Toru Odate:** Validation, Investigation. **Kazunari Kasai:** Validation. **Tomohiro Inoue:** Validation. **Kunio Mochizuki:** Validation. **Naoki Oishi:** Writing – review & editing. **Tetsuo Kondo:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests. Masataka Kawai has patent pending to University of Yamanashi.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.combiomed.2024.109149>.

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